Chapter 3: Environmental and Social Determinants of Cognitive Health in Puente Piedra, Peru

Gabrielle Husted

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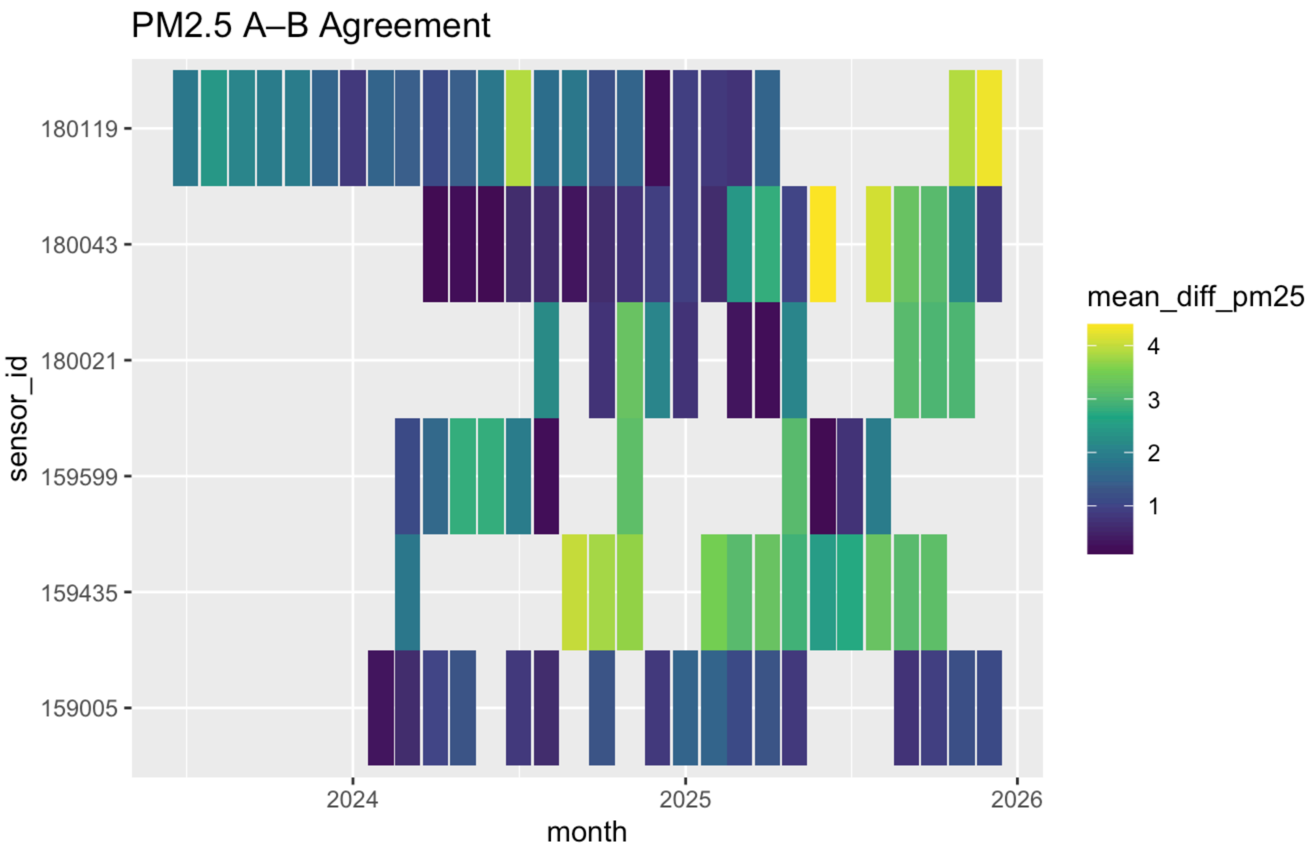
0.1 Download Historical Purple Air Data via Data Download Tool

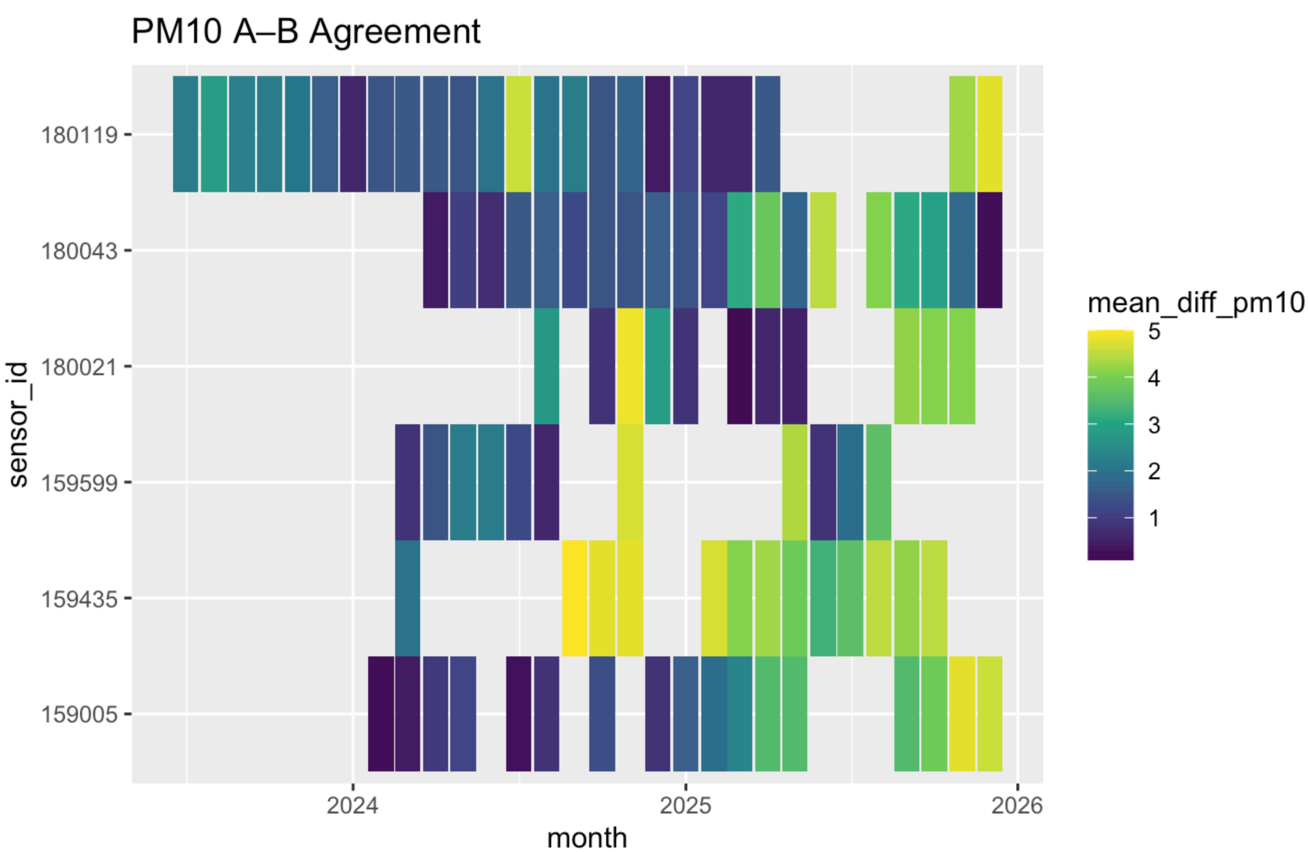
Sensor history: https://api.purpleair.com/#api-sensors-get-sensor-history (https://api.purpleair.com/#api-sensors-get-sensor-history)

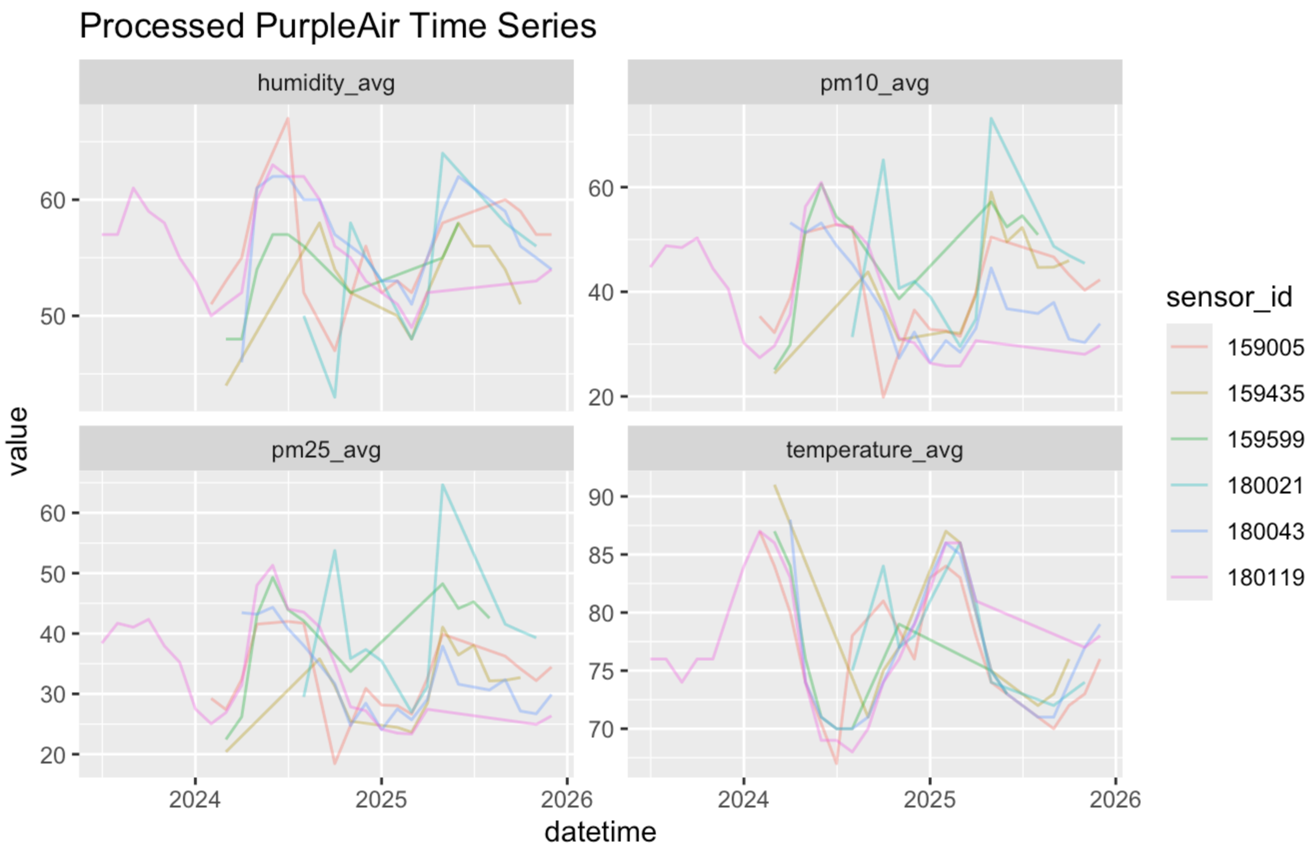
Fields: pm10.0_atm_a,pm10.0_atm_b,pm10.0_atm,voc_a,voc_b,voc,humidity,humidity_a,humidity_b, temperature,temperature_a,temperature_b,pm2.5_atm_a,pm2.5_atm_b,pm2.5_atm

Date range: 22-01-01 to 26-01-10

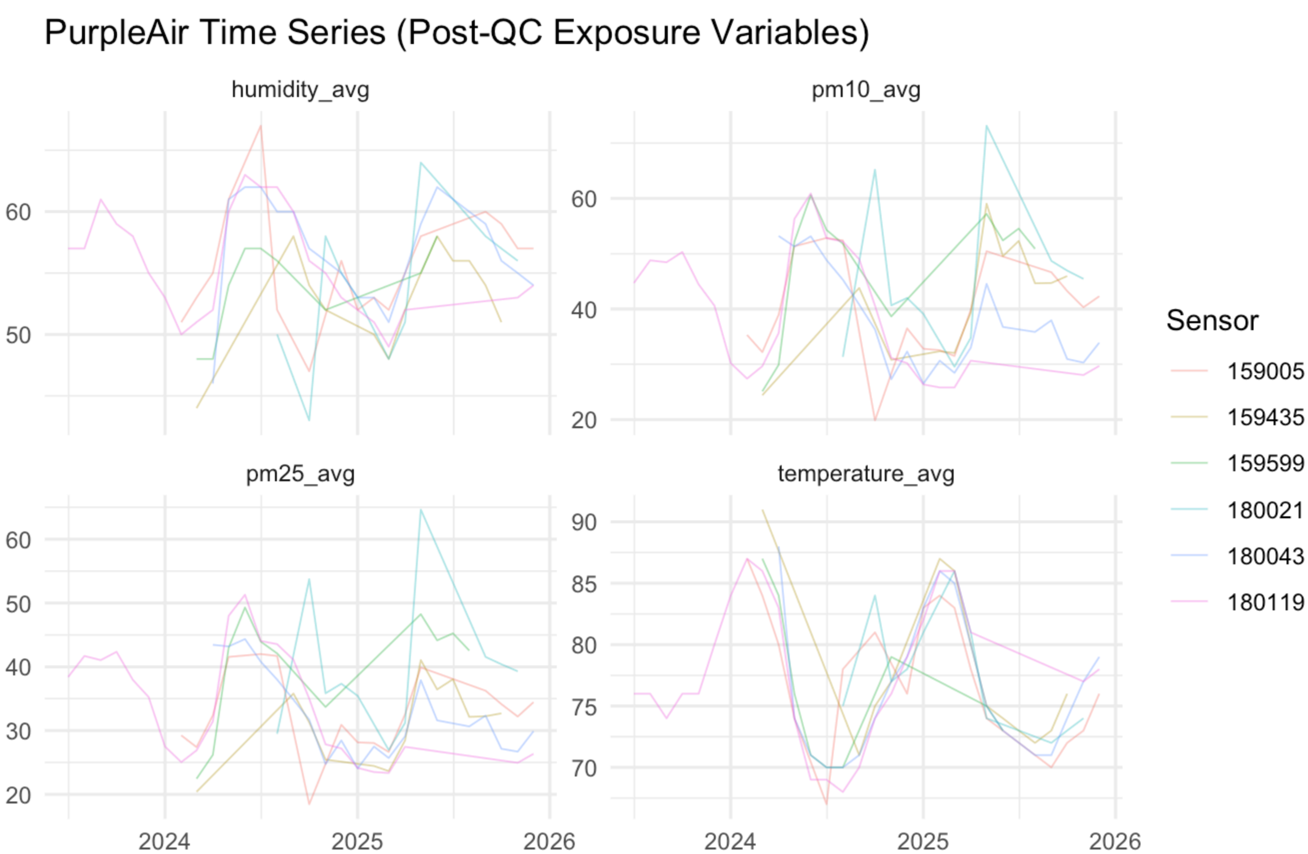
1-month average

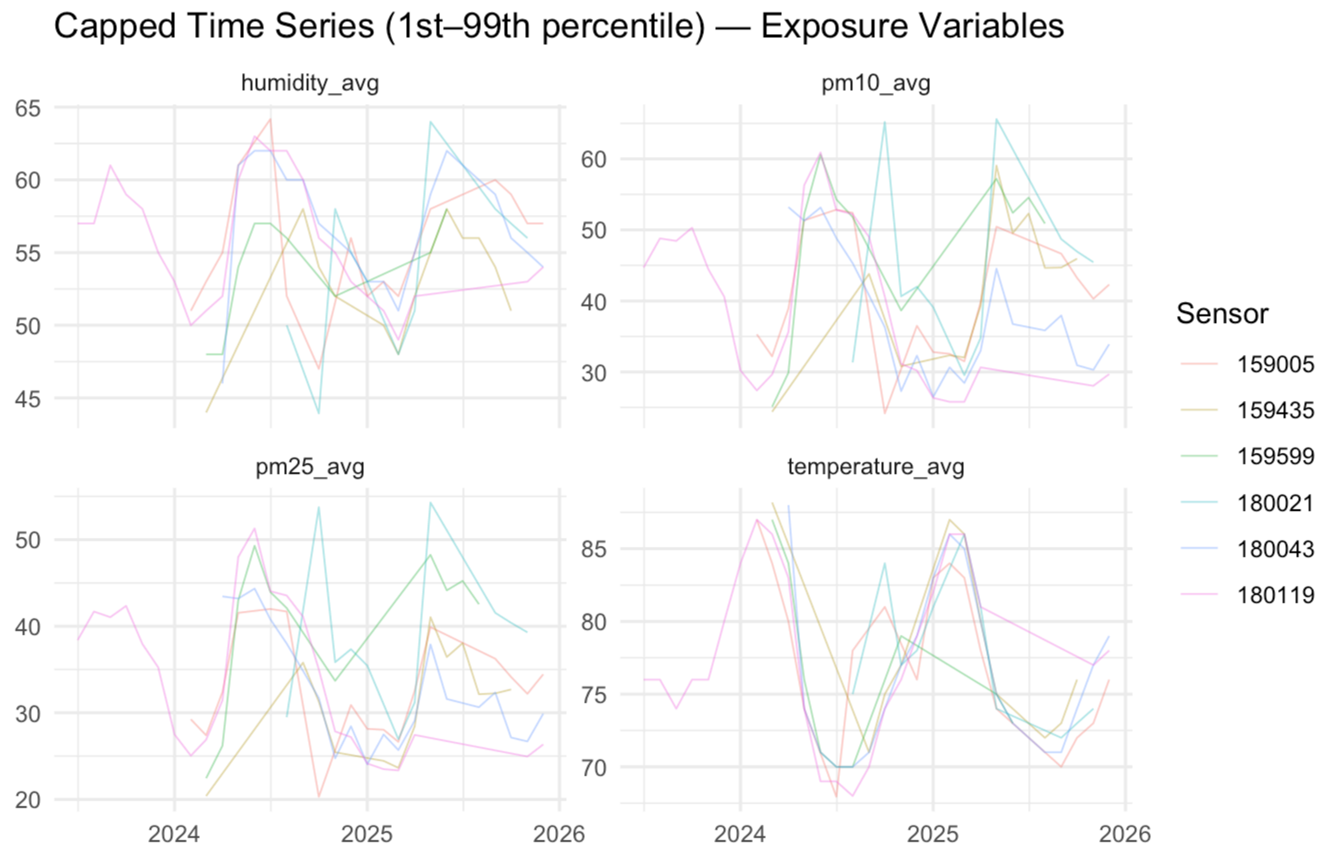






0.2 Visualizations: Processed Exposure Time Series






```
## # A tibble: 4 × 9
##   variable              n  mean    sd median   p25   p75   min   max
##   <chr>             <int> <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Humidity (%)          6  54.8 1.46    54.7  53.8  55.6  52.9  56.8
## 2 PM10 (µg/m³)          6  41.9 3.89    40.6  39.4  44.2  37.9  48.0
## 3 PM2.5 (µg/m³)         6  35.0 3.89    33.3  32.6  38.2  30.9  40.1
## 4 Temperature (°F)      6  77.2 0.889   77.7  76.7  77.7  75.7  77.8
```

0.3 Adding location of sensor

Unable to get this via the data download tool, need to use the API directly

```
##
## FALSE
##     6
```

```
## # A tibble: 6 × 11
##   sensor_id mean_pm25 mean_pm10 mean_humidity mean_temperature abs_thresh
##   <fct>         <dbl>     <dbl>         <dbl>            <dbl>      <dbl>
## 1 159005         32.7      39.9          55.6             77.6          5
## 2 159435         30.9      41.3          52.9             77.7          5
## 3 159599         40.1      48.0          53.8             75.7          5
## 4 180021         39.6      45.2          53.9             77.7          5
## 5 180043         32.6      37.9          56.8             76.4          5
## 6 180119         34.0      39.2          55.6             77.8          5
## # ℹ 5 more variables: rel_thresh <dbl>, sensor_id_clean <dbl>,
## #   sensor_name <chr>, latitude <dbl>, longitude <dbl>
```

```
## Deleting source `/Users/gabriellebenoit/Documents/GitHub/peru_air/purpleair__all_sens
ors.gpkg' using driver `GPKG'
## Writing layer `purpleair__all_sensors' to data source
##   `/Users/gabriellebenoit/Documents/GitHub/peru_air/purpleair__all_sensors.gpkg' usin
g driver `GPKG'
## Writing 6 features with 2 fields and geometry type Point.
```

0.4 Socio-demographic Data

New map, numbered by section. Final version of the map (https://www.google.com/maps/d/edit?mid=1ZxIT6hY4dc9rtQl3oUCLNVQoDkfRK5w&usp=sharing) includes sector 5.

Confirmed (as of March 25th, 2026) working dataset: “Base de datos (Monica)_09.16.2024_MN edits 09.19.2024 copy 3.xlsx”

0.5 Primary variables of interest

Cognitive scores: - ACE - RUADS - PFAQ - Dementia - Dementia AM - Dementia AJ - Dementia AJ ALT - deterioro cognitivo

Exposure-related: - pollution (pm2.5, pm10) - temperature - humidity

Geography: (account for clustering) - Distrito - Sector - Conglomerado/pin

Core standard confounders: - age - sex - education - occupation

Health-related: - bmi - glucose_lev - hypertension

Socioeconomic: - overcrowding - health_insurance - civil_stat

0.6 Using the google map data

Only 4 unmatched out of 1,099 rows.. negligible. Will proceed with the join. Those 4 participants simply won’t have coordinates and will be dropped from the spatial analysis.

Many-to-many warning – makes sense, as it is expected that multiple survey participants will live in the same conglomerate (same pin/location point), so one KML point matches many survey rows. Epected for cluster-based sampling.

```
## Driver: KML
## Available layers:
##   layer_name geometry_type features fields crs_name
## 1   Sector 3      3D Point      264      2   WGS 84
## 2   Sector 1      3D Point      198      2   WGS 84
## 3   Sector 4      3D Point      200      2   WGS 84
## 4   Sector 2      3D Point      185      2   WGS 84
## 5   Sector 5      3D Point      259      2   WGS 84
```

```
## [1] 1106
```

```
## [1] "Name"        "Description" "geometry"
```

```
## Simple feature collection with 6 features and 2 fields
## Geometry type: POINT
## Dimension:     XYZ
## Bounding box:  xmin: -77.09736 ymin: -11.8496 xmax: -77.0945 ymax: -11.84723
## z_range:       zmin: 0 zmax: 0
## Geodetic CRS:  WGS 84
##   Name Description                       geometry
## 1  407             POINT Z (-77.09736 -11.8472...
## 2  409             POINT Z (-77.09628 -11.8477...
## 3  415             POINT Z (-77.09686 -11.8482 0)
## 4  417             POINT Z (-77.09519 -11.8496 0)
## 5  418             POINT Z (-77.09572 -11.8490...
## 6  420             POINT Z (-77.0945 -11.84954 0)
```

```
##
## FALSE  TRUE
##     4  1095
```

```
##
## Sector 1 Sector 3
##        1        3
```

```
## [1] 1989
```

```
## Deleting source `/Users/gabriellebenoit/Documents/GitHub/peru_air/spatial_survey_dat
a.gpkg' using driver `GPKG'
## Writing layer `spatial_survey_data' to data source
##   `/Users/gabriellebenoit/Documents/GitHub/peru_air/spatial_survey_data.gpkg' using d
river `GPKG'
## Writing 1989 features with 25 fields and geometry type 3D Point.
```

0.7 Spatial Join for All Variables

```
## Simple feature collection with 6 features and 6 fields
## Geometry type: POINT
## Dimension:     XYZ
## Bounding box:  xmin: 271508.3 ymin: 8689420 xmax: 271626.9 ymax: 8689480
## z_range:       zmin: 0 zmax: 0
## Projected CRS: WGS 84 / UTM zone 18S
##   conglomerado_chr         nearest_sensor_name exposure_pm25 exposure_pm10
## 1              407 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
## 2              407 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
## 3              409 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
## 4              409 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
## 5              409 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
## 6              409 GEOHealth_MiPeru_Chavinillo      32.70588      39.94412
##   exposure_temp exposure_humidity                     geometry
## 1      77.64706          55.58824 POINT Z (271508.3 8689480 0)
## 2      77.64706          55.58824 POINT Z (271508.3 8689480 0)
## 3      77.64706          55.58824 POINT Z (271626.9 8689420 0)
## 4      77.64706          55.58824 POINT Z (271626.9 8689420 0)
## 5      77.64706          55.58824 POINT Z (271626.9 8689420 0)
## 6      77.64706          55.58824 POINT Z (271626.9 8689420 0)
```

```
## Deleting source `/Users/gabriellebenoit/Documents/GitHub/peru_air/spatial_data_final.
gpkg' using driver `GPKG'
## Writing layer `spatial_data_final' to data source
##   `/Users/gabriellebenoit/Documents/GitHub/peru_air/spatial_data_final.gpkg' using dr
iver `GPKG'
## Writing 1989 features with 31 fields and geometry type 3D Point.
```

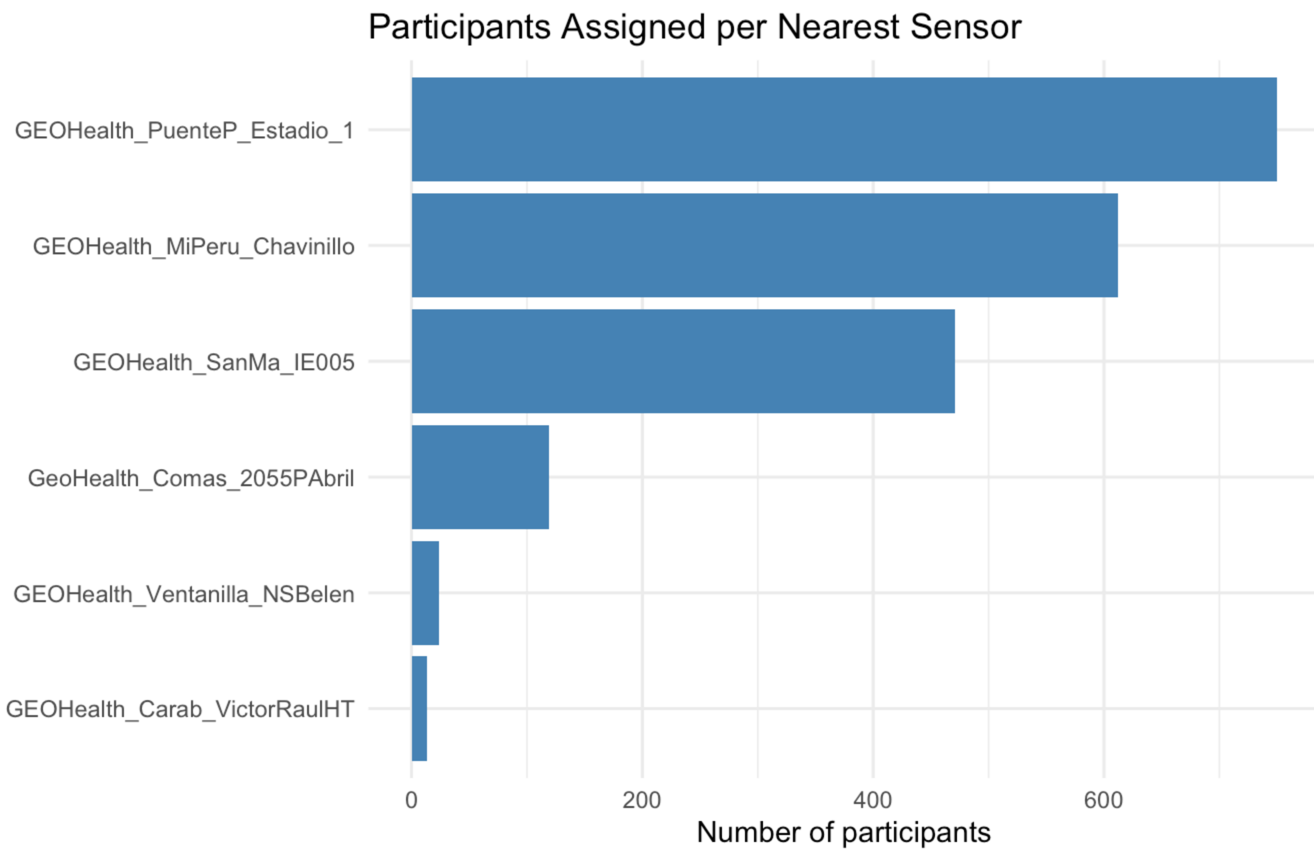
0.8 Explore which sensors were used, how many participants were assigned to it

A few things to note: - 180119 and 159005 cover the bulk of participants (750 and 612) — results will be heavily influenced by these two sensors - 159435 and 180021 have very few participants assigned (24 and 13) — these are likely on the geographic periphery of the study area

Limitations: exposure assignment was constrained to 6 sensors, with the majority of participants assigned to just 2 sensors — which reduces spatial variability in exposure estimates.

1 Load Data

```
##   nearest_sensor_id          nearest_sensor_name   n
## 1            180119  GEOHealth_PuenteP_Estadio_1 750
## 2            159005  GEOHealth_MiPeru_Chavinillo 612
## 3            180043        GEOHealth_SanMa_IE005 471
## 4            159599   GeoHealth_Comas_2055PAbril 119
## 5            159435 GEOHealth_Ventanilla_NSBelen  24
## 6            180021 GEOHealth_Carab_VictorRaulHT  13
```



1.1 Examine contextual variables – clean

```
##    Var1 Freq
## 1     0   97
## 2     1   31
## 3    10   54
## 4    11  549
## 5    12  101
## 6    13   77
## 7    14  149
## 8  14.5    1
## 9    15   30
## 10   16   79
## 11   17    6
## 12   18    9
## 13   19   10
## 14    2   40
## 15   20    2
## 16   23    1
## 17    3   60
## 18    4   44
## 19    5   86
## 20    6  290
## 21    7   39
## 22    8   52
## 23    9  110
## 24 <NA>   72
```

```
##
##                           0    1
##   none                   57   29
##   primary               391  135
##   secondary_or_higher 1136  115
```

```
##      civil_stat   n
## 1   Conviviente 645
## 2     Casado(a) 552
## 3    Soltero(a) 408
## 4   Separado(a) 141
## 5      Viudo(a) 138
## 6          <NA>  73
## 7 Divorciado(a)  32
```

```
##          occupation   n
## 1    Amo(a) de casa 634
## 2  Auto-empleado(a) 446
## 3 Sin empleo actual 327
## 4       Empleado(a) 297
## 5       Jubilado(a) 114
## 6         Freelance  94
## 7              <NA>  75
## 8        Estudiante   2
```

```
##
##         partnered            single separated_widowed              <NA>
##              1197               408               311                73
```

```
##
##     employed    homemaker not_employed         <NA>
##          837          634          443           75
```

```
##
## --- education_cat ---
##
##                          0    1
##   none                  57   29
##   primary              387  135
##   secondary_or_higher 1134  112
##
## --- civil_stat_cat ---
##
##                       0   1
##   partnered         979 182
##   single            334  58
##   separated_widowed 265  36
##
## --- occupation_cat ---
##
##                 0   1
##   employed     715 101
##   homemaker    482 124
##   not_employed 381  51
##
## --- htn ---
##
##                          0    1
##   HTA diastólica        87   19
##   HTA sistodiastólica  185   33
##   HTA sistólica        154   10
##   Sin hipertensión    1152  214
##
## --- native_lang ---
##
##                       0    1
##   Castellano       1306  176
##   Quechua o aimara  272  100
##
## --- health_ins ---
##
##                                    0   1
##   Entidad prestadora de salud     11   1
##   Essalud                        349  32
##   Fuerzas armadas o policiales    46   0
##   No está afiliado               217  45
##   No sabe                         35  22
##   Seguro integral de salud (SIS) 888 175
##   Seguro privado de salud         32   1
##
## --- chronic_dis ---
##
##        0   1
##   No 982 190
##   Sí 596  86
```

```
##
## --- sex ---
##
##                0    1
##   Femenino  1067  242
##   Masculino  511   34
```

```
##
##              0   1
##   SIS       890 178
##   Essalud   350  32
##   uninsured 220  45
##   other     124  24
```

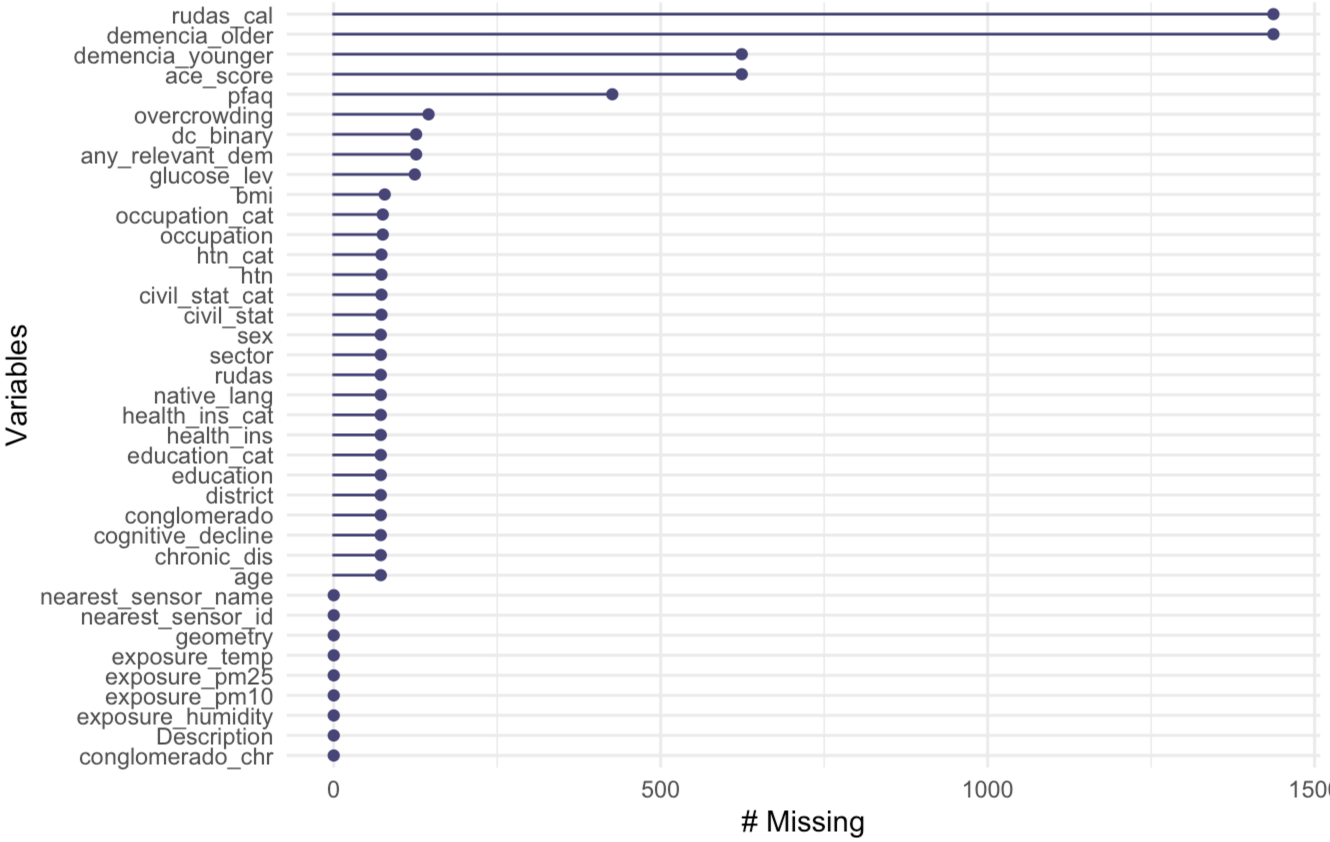
```
##
##                          0    1
##   HTA diastólica        87   19
##   HTA sistodiastólica  185   33
##   HTA sistólica        154   10
##   Sin hipertensión    1152  214
```

```
##
##                    0    1
##   none         1154  217
##   hypertension  429   62
```

```
## [1] "numeric"
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   15.82   25.51   28.32   28.85   31.42   51.85      78
```

1.2 Check missingness




```
##
##    0    1
## 1584  279
```

```
## [1] 1854
```

```
##   exposure_pm25   n
## 1      30.93462  24
## 2      32.60750 471
## 3      32.70588 612
## 4      33.97500 750
## 5      39.63182  13
## 6      40.08182 119
```

1.3 Create standardized versions of PM 2.5, PM 10, temperature, and humidity

and quartile for PM 2.5

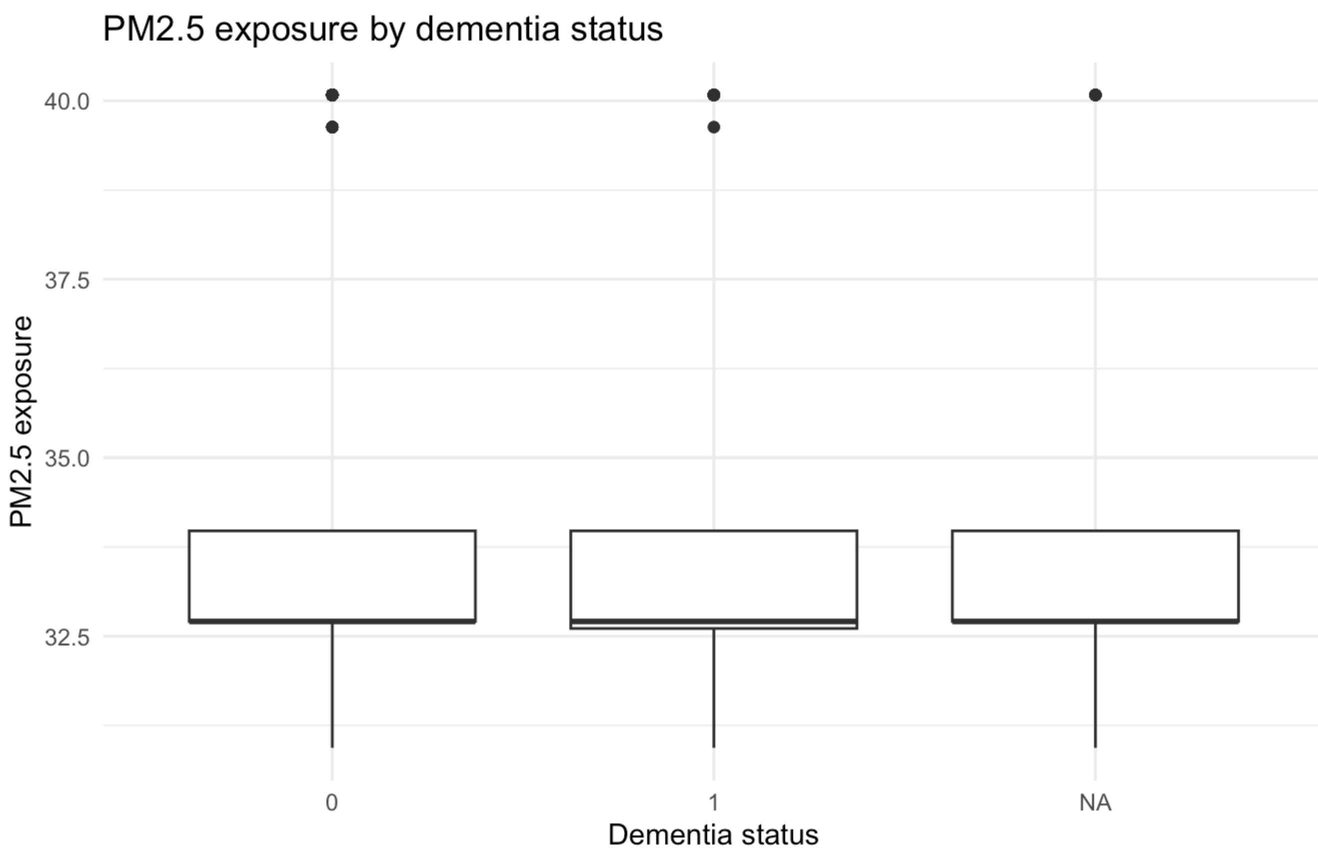
Interpretation: SD models: OR = dementia odds per 1 SD increase in exposure Quartile models: OR = dementia odds relative to lowest exposure group

```
## [1] "numeric"
```

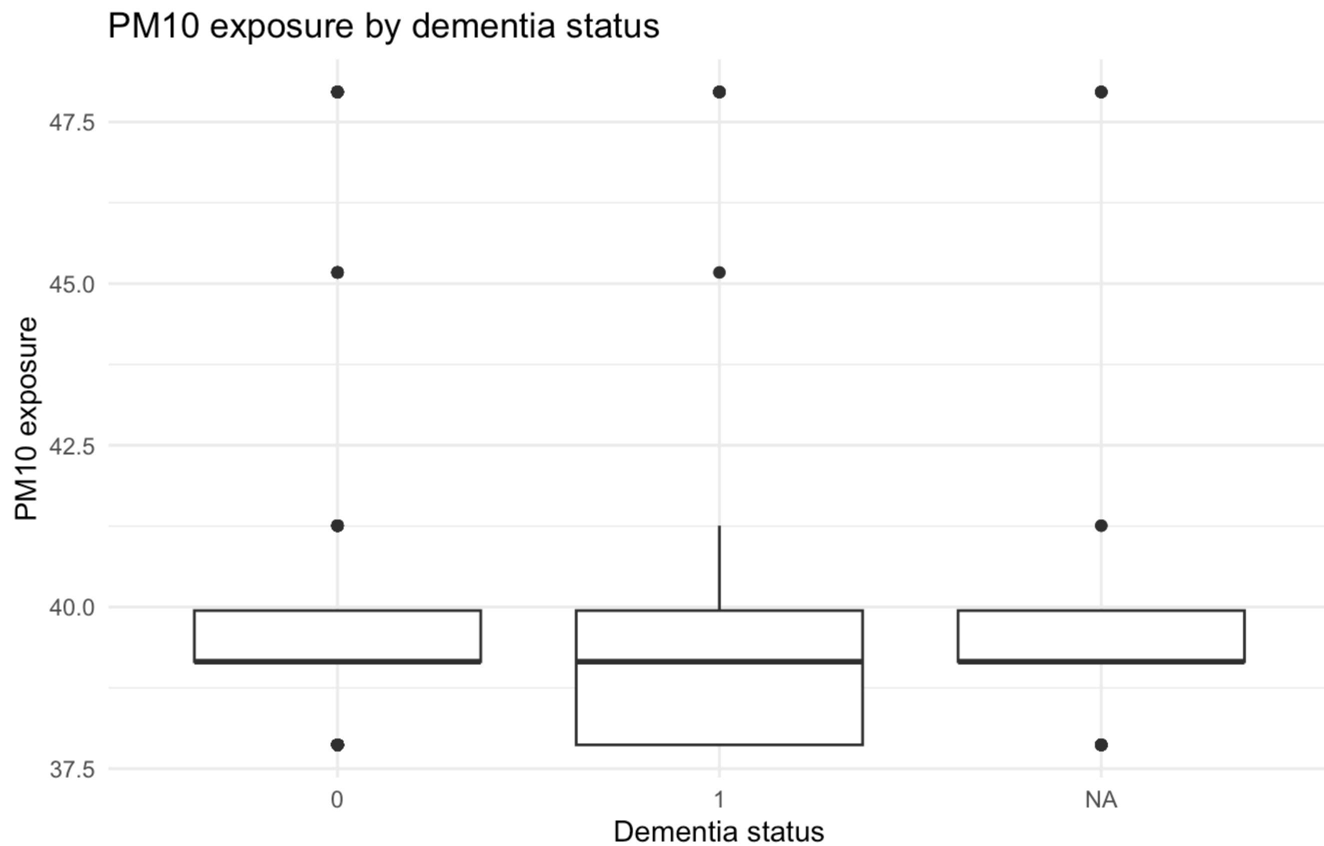
1.4 Descriptive Exploration

```
## # A tibble: 3 × 6
##   dc_binary     n pm25_mean pm25_sd pm10_mean pm10_sd
##       <dbl> <int>     <dbl>   <dbl>     <dbl>   <dbl>
## 1         0  1584      33.6    1.85      39.7    2.30
## 2         1   279      33.6    1.78      39.6    2.22
## 3        NA   126      33.6    1.84      39.8    2.29
```

boxplots # pm 2.5



2 pm 10



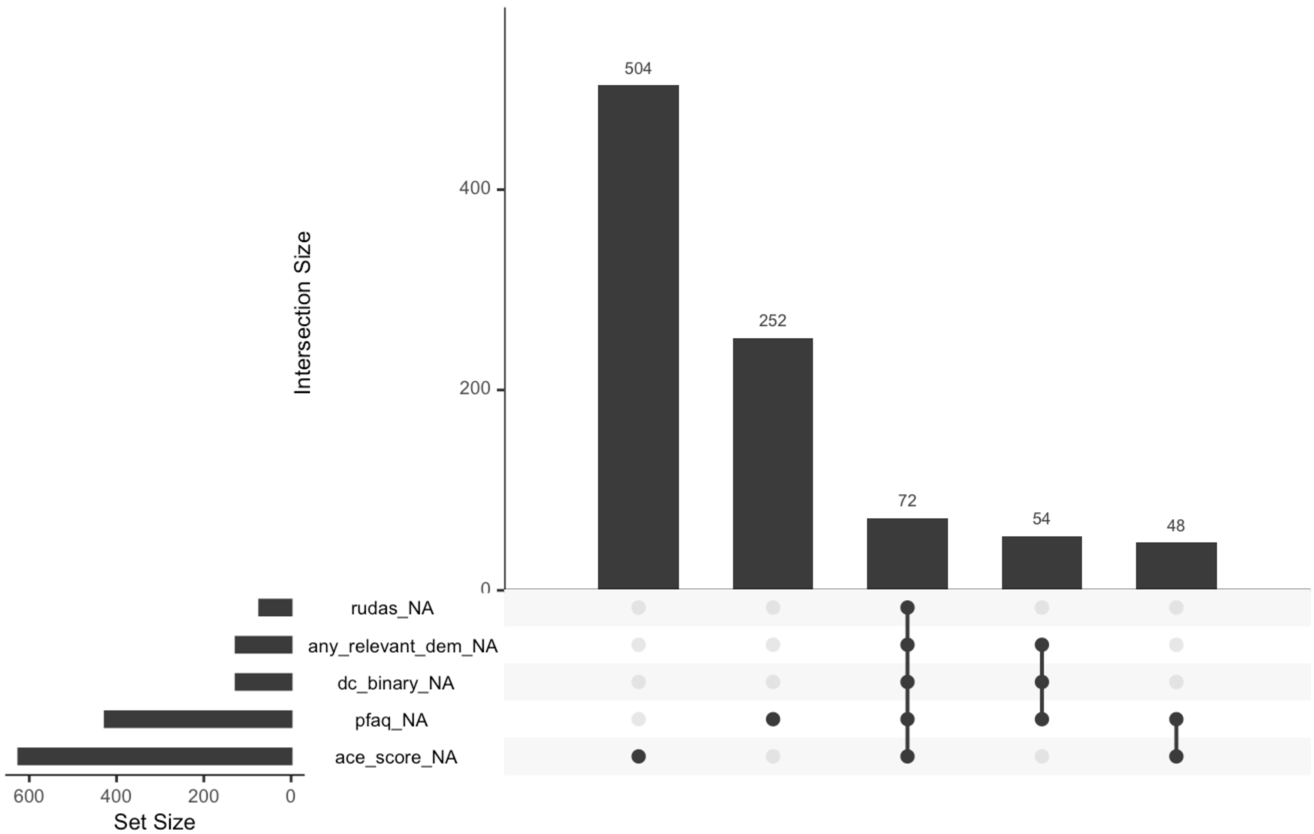
2.1 Analysis (Thursday, April 9)

1. dc_binary or any_relevant_dem as primary outcome
2. stratified analyses for older and younger adults with their continuous scores as secondary outcome

determine how many NAs there are for any_relevant dementia, dc_binary, rudas, ace, pfaq # how do they or do they not overlap?

```
## na_per_row
##    0    1    2    3    5
## 1059  756   48   54   72
```

```
## any_relevant_dem        dc_binary            rudas        ace_score
##              126              126               72              624
##             pfaq         geometry
##              426                0
```



Delete observations that are missing all five variables because they need at least one of those measures to be able to be in our analyses meaningfully.

```
## [1] 1917
```

```
##   n_rudas n_ace n_pfaq n_dc
## 1    1917  1365   1563 1863
```

```
## # A tibble: 2 × 4
##   took_rudas took_ace mean_age     n
##   <lgl>      <lgl>       <dbl> <int>
## 1 TRUE       FALSE        69.1   552
## 2 TRUE       TRUE         43.2  1365
```

```
## # A tibble: 2 × 5
##   took_ace min_age max_age mean_age     n
##   <lgl>      <dbl>   <dbl>    <dbl> <int>
## 1 FALSE         60      95     69.1   552
## 2 TRUE          30      59     43.2  1365
```

2.2 Primary analysis: full sample

```
## dc_binary ~ exposure + age + sex
```

2.3 Secondary analysis: older adults (60+)

```
## as.numeric(rudas) ~ exposure + age + sex
```

2.4 Secondary analysis: younger adults (30-59):

```
## as.numeric(ace_score) ~ exposure + age + sex
```

2.5 Create the age groups:

2.6 Prep

2.7 Model 1

Full sample, binary outcome

2.8 Model 2

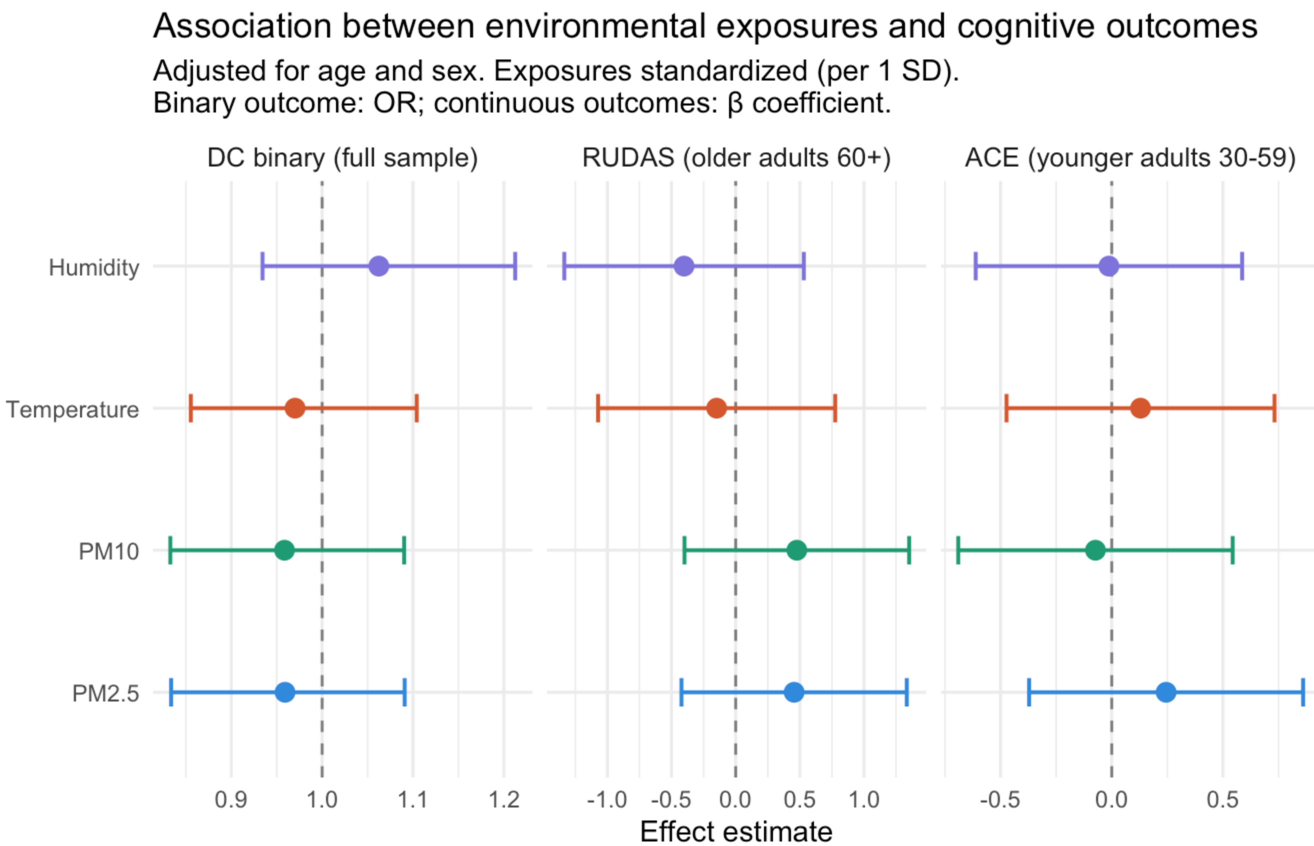
Older adults (60+), RUDAS

2.9 Model 3

Younger adults (30-59), ACE

2.10 Combine

All results - binary results are ORs - linear results are beta coefficients - they are in separate facets so the scales don't mix




```
## # A tibble: 4 × 5
##   exposure    estimate conf.low conf.high p.value
##   <chr>          <dbl>    <dbl>     <dbl>   <dbl>
## 1 PM2.5          0.456   -0.423     1.33    0.309
## 2 PM10           0.477   -0.399     1.35    0.285
## 3 Temperature   -0.148   -1.07      0.776   0.753
## 4 Humidity      -0.402   -1.34      0.531   0.397
```

2.11 Sensor zone comparison table to show descriptively whether zones with higher PM2.5 tend to have worse cognitive scores

No dose-response pattern for PM2.5 → cognitive decline. Looking at the four sensors with adequate sample size (ignore Ventanilla n=22 and Carab n=13 — too small): The highest PM2.5 zone (Comas) has the lowest % DC and highest RUDAS scores. This is an "ecological confounding" artifact made visible — Comas residents are younger, more educated, or otherwise healthier despite higher pollution. The sensor zones are not exchangeable populations.

This table is a valuable finding — it demonstrates why naive ecological exposure assignment produces misleading results, and motivates future work with better exposure characterization; a methodological lesson

```
## # A tibble: 6 × 6
##   nearest_sensor_name           pm25 pct_dc mean_rudas mean_ace     n
##   <chr>                        <dbl>  <dbl>      <dbl>    <dbl> <int>
## 1 GEOHealth_Ventanilla_NSBelen  30.9   14.3       3.45     77.9    22
## 2 GEOHealth_SanMa_IE005         32.6   16.6       6.80     78.1   456
## 3 GEOHealth_MiPeru_Chavinillo   32.7   15.1       6.85     77.6   587
## 4 GEOHealth_PuenteP_Estadio_1   34.0   14.1       7.84     79.0   725
## 5 GEOHealth_Carab_VictorRaulHT  39.6   15.4       5.38     78.4    13
## 6 GeoHealth_Comas_2055PAbril    40.1   13.5      10.4      78.0   114
```

2.12 Highways

The study area runs along what looks like a major highway corridor (the blue trunk road running north-south). Our participant points are clustered along and near this road. The red/pink dementia cases don’t obviously cluster in one spot, but proximity to that trunk road varies meaningfully across participants — unlike the PM2.5 values which barely vary at all.

```
## Reading layer `highways_export' from data source
##   `/Users/gabriellebenoit/Documents/GitHub/peru_air/highways_export.gpkg'
##   using driver `GPKG'
## Simple feature collection with 345 features and 24 fields
## Geometry type: LINESTRING
## Dimension:     XY
## Bounding box:  xmin: 264542.2 ymin: 8681310 xmax: 277584.6 ymax: 8692089
## Projected CRS: WGS 84 / UTM zone 18S
```

```
##  [1] "osm_id"          "name"            "alt_name"        "bridge"
##  [5] "destination"     "destination.ref" "foot"            "highway"
##  [9] "junction"        "lane_markings"   "lanes"           "layer"
## [13] "maxheight"       "maxspeed"        "motorroad"       "noname"
## [17] "note"            "old_name"        "oneway"          "placement"
## [21] "ref"             "source"          "surface"         "turn.lanes"
## [25] "geom"
```

```
## < table of extent 0 >
```

```
## [1] 345
```

```
##
##   primary secondary     trunk
##        22       240        83
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##    9.293  331.501  754.896 1240.405 2053.607 4612.723
```

Meaningful continuous variation from 9 meters to 4.6 km. - Median 755m — half of the participants live within ~750m of a major road - Some very close (9m) — essentially on the road - Max 4.6km — good spread at the far end - Right-skewed (mean 1,240m >> median 755m) — confirms the need to log-transform

```
## # A tibble: 4 × 7
##   term          estimate std.error statistic  p.value conf.low conf.high
##   <chr>            <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)      0.239   0.439       -3.26 1.11e- 3   0.0999     0.560
## 2 log_dist_road    1.07    0.0523       1.29 1.95e- 1   0.967      1.19
## 3 age_n            0.990   0.00491     -2.04 4.11e- 2   0.980      1.00
## 4 sexMasculino     0.305   0.192       -6.19 6.17e-10   0.206      0.438
```

```
## # A tibble: 4 × 7
##   term          estimate std.error statistic p.value conf.low conf.high
##   <chr>            <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
## 1 (Intercept)    8.71       5.50     1.58      0.114   -2.09     19.5
## 2 log_dist_road -0.112      0.368   -0.304     0.762   -0.835     0.611
## 3 age_n         -0.0101     0.0699  -0.144     0.885   -0.147     0.127
## 4 sexMasculino  -0.00838    0.975   -0.00859   0.993   -1.92      1.91
```

```
## # A tibble: 4 × 7
##   term          estimate std.error statistic   p.value conf.low conf.high
##   <chr>            <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
## 1 (Intercept)     92.5      2.30       40.2  1.28e-233   88.0      97.0
## 2 log_dist_road   -0.562    0.235      -2.40 1.67e-  2   -1.02     -0.102
## 3 age_n           -0.269    0.0358     -7.51 1.03e- 13   -0.339    -0.199
## 4 sexMasculino     4.69     0.720       6.51 1.02e- 10    3.28      6.10
```

```
## # A tibble: 2 × 5
##   near_road_500m pct_dc mean_rudas mean_ace     n
##   <lgl>           <dbl>      <dbl>    <dbl> <int>
## 1 FALSE            16.2       7.13     77.8  1250
## 2 TRUE             12.7       7.84     79.2   667
```

This is the most interesting result yet. Participants within 500m of a major road have lower rates of cognitive decline (12.7% vs 16.2%) and higher cognitive scores (RUDAS 7.84 vs 7.13, ACE 79.2 vs 77.8) — the opposite of what you'd expect if road proximity is harmful. This is the same ecological confounding pattern observed with PM2.5 — the Comas sensor (highest PM2.5) also had the best cognitive scores. It's telling a consistent story: participants living closer to major roads in this study area are systematically different in ways that protect cognition — likely younger, more educated, better access to healthcare, higher SES. Urban proximity in a peri-urban Lima context may mean better services, not worse health.

2.13 Regression Models adjusted for age and sex

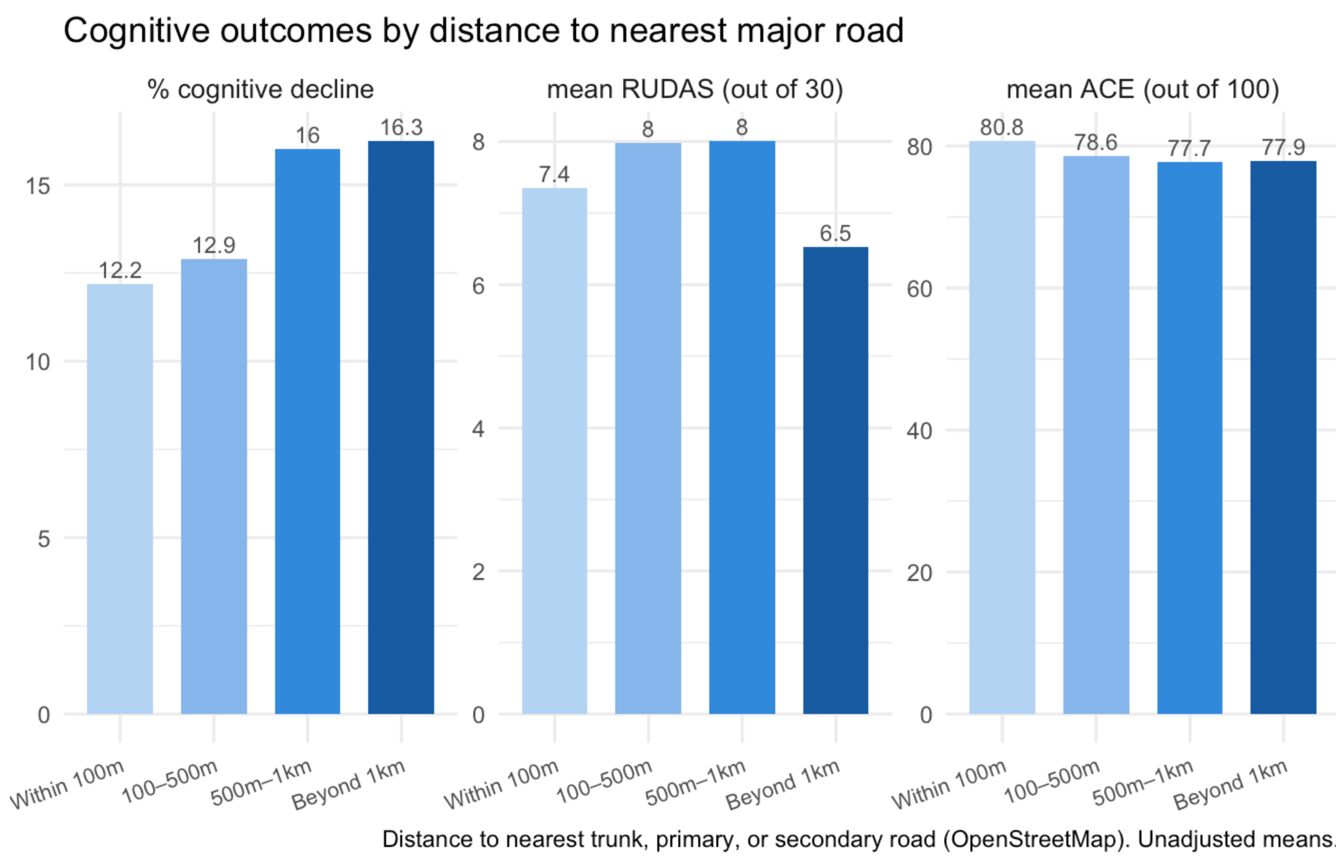
```
## # A tibble: 1 × 5
##   term          estimate conf.low conf.high p.value
##   <chr>            <dbl>    <dbl>     <dbl>   <dbl>
## 1 log_dist_road     1.07    0.967      1.19   0.195
```

```
## # A tibble: 1 × 5
##   term          estimate conf.low conf.high p.value
##   <chr>            <dbl>    <dbl>     <dbl>   <dbl>
## 1 log_dist_road   -0.112   -0.835     0.611   0.762
```

```
## # A tibble: 1 × 5
##   term          estimate conf.low conf.high p.value
##   <chr>            <dbl>    <dbl>     <dbl>   <dbl>
## 1 log_dist_road   -0.562    -1.02    -0.102  0.0167
```

2.14 Figure

```
## # A tibble: 4 × 5
##   buffer_zone pct_dc mean_rudas mean_ace     n
##   <fct>        <dbl>      <dbl>    <dbl> <int>
## 1 Within 100m   12.2       7.35     80.8   159
## 2 100–500m      12.9       7.99     78.6   508
## 3 500m–1km      16.0       8.01     77.7   506
## 4 Beyond 1km    16.3       6.53     77.9   744
```



The only significant finding is ACE in younger adults. Binary and RUDAS are null. This is a coherent and interpretable pattern

Only younger adults? possibly because ACE is a more sensitive, higher-ceiling instrument than RUDAS. Younger adults (mean age 43) have more cognitive reserve and more variance in scores to detect — so subtle environmental effects show up there first. Older adults' RUDAS scores may already reflect age-related decline that swamps any road proximity effect.

Implications of the ACE findings: Among 30–59 year olds, each unit increase in log-distance from a major road is associated with a 0.56 point decrease in ACE score. Given ACE is scored out of 100, that's modest but statistically present. The direction — farther from roads = worse cognition — points to urban access rather than pollution as the operative mechanism in this setting.

2.15 The *storyline* for the chapter:

1. PM2.5 sensor zones — null, insufficient spatial variability
2. Distance to road — significant for ACE in younger adults, counterintuitive direction
3. Interpretation — urbanization-health paradox in peri-urban Lima, road proximity as access proxy

Implication — future studies need personal, granular exposure monitors + SES adjustment to disentangle pollution from access